Osama Abdul Qader

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Technical Skills

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript
Frameworks: React, Next.js, Tailwind, Bootstrap, Django, DjangoRestAPI
Developer Tools: Git, Docker, Redis, Google Cloud Platform, VS Code, Grafana
Libraries: pandas, NumPy, Matplotlib

Projects

LedgerLock | *Python, Django, PostgreSQL, Redis* | GitHub — Oct 2026 – Present

- Engineered high-concurrency API endpoints and designed an optimized database schema for a double-entry ledger in Django and PostgreSQL, utilizing row-level locking (FOR UPDATE) to guarantee ACID compliance and completely eliminate balance race conditions under high traffic.
- Implemented distributed idempotency management via atomic Redis Lua scripting, effectively blocking duplicate transactions, double-spends, and replay attacks across parallel API calls.
- Integrated Locust load-testing scripts to evaluate transaction throughput and verify thread safety under high concurrent traffic.
- Containerized the application and infrastructure using Docker Compose for seamless orchestration of the Django app, PostgreSQL database, and Redis cache.

TaskEnd | *Python, Django, PostgreSQL, Redis* | Live Link | GitHub — June 2026 – September 2026

- Benchmarked the deployed application using Locust, recording 812 requests with zero failures at 20 simulated users; tested 100-user load and identified performance bottlenecks.
- Designed the PostgreSQL database schema and REST API endpoints for a full-stack task management application built with Django and Django REST Framework.
- Implemented authentication, Google OAuth, user-specific task management, and CRUD operations with filtering, completion tracking, and status management; added rate limiting for endpoint protection.
- Containerized services using Docker Compose with PostgreSQL, Redis, Gunicorn, and Nginx; deployed on Render with Git-based continuous deployment.

Open Source Collaboration

PostgreSQL — August 2026 – September 2026

GitHub | *Commit Merged*

- Contributed an upstream fix to PostgreSQL to reject REPACK (ANALYZE) inside transaction blocks and PL/pgSQL execution contexts.
- Added regression tests covering transaction-block and function/procedure execution scenarios.
- Updated PostgreSQL documentation to reflect the transaction restrictions for REPACK (ANALYZE).

Markdown-Reader — Jul. 2025 – Aug. 2025

GitHub | *Pull Request Merged*

- Contributed and merged a pull request after maintainer review, using Git and GitHub for version control and code review.

Education

Sphoorthy Engineering College — Hyderabad, Telangana
Bachelor of Technology in Computer Science Engineering — *September 2023 – May 2026*

Sphoorthy Engineering College — Hyderabad, Telangana
Polytechnic Diploma in Computer Science Engineering — *November 2020 – May 2023*